

final answer. This finding is promising, suggesting that enhancing the intermediate PVD representation, even with a fixed reasoning model, can effectively boost downstream task performance.

4.2 Interpretable Error Analysis

The improved interpretability, resulting from PVD’s disentangled perception and reasoning, allows us to conduct an in-depth analysis of the failure modes of VDLM. We find that both the perception step (SVG-to-PVD) and the reasoning step (PVD-to-answer) can contribute to errors. On tasks that require complex multistep reasoning, such as Maze Solving, reasoning errors become more prevalent; otherwise, perception errors most directly contribute to poor performance. Details and illustrative examples of these errors are provided in **Appendix B**, along with a distribution of perception and reasoning errors from human analyses. The prevalent error types for both perception and reasoning steps are summarized as follows.

Common perception errors include failures in faithfully perceiving novel shapes that are not covered by or cannot be composed within the PVD ontology, and failures in capturing intentional constraints between primitives, such as a line exactly segmenting a circle, due to the random nature of the data generation on the positioning of objects. In Table 5, we show that the proposed augmentation during synthetic data generation improves PVD perception. Common reasoning errors over the PVD perception include failures in discovering intentional constraints without being explicitly asked, such as automatically recognizing that a rhombus is not the same concept as a general quadrilateral; failure in handling ambiguous instructions; and failure in complex multi-step reasoning tasks like solving mazes.

5 Related Work

Visual shortcomings in large multimodal models. While state-of-the-art LMMs achieve strong performance on existing multimodal benchmarks (Goyal et al., 2017; Fu et al., 2023; Liu et al., 2023b;d; Yu et al., 2023; Li et al., 2023a), which primarily focus on natural images, recent work (Lu et al., 2023; Yue et al., 2023; Huang et al., 2023; Zhou et al., 2023a; Hsu et al., 2022; Gao et al., 2023) has shown that they struggle with charts, geometric diagrams, and abstract scenes. This observation aligns with recent studies investigating visual shortcomings in LMMs. Tong et al. (2024) suggests that current LMMs struggle with visual details because the image-text contrastive pretraining of the CLIP visual backbone does not encourage the preservation of fine-grained visual features, such as orientation and quantity. To address this issue, recent studies have either leveraged the mixture-of-experts approach (Tong et al., 2024; Fan et al., 2024; Lu et al., 2024; Jain et al., 2023b), incorporating various types of vision encoders, such as SAM (Kirillov et al., 2023), DINOv2 (Oquab et al., 2023), or introduced auxiliary losses that emphasize local details during multimodal pretraining McKinzie et al. (2024); Bica et al. (2024); Varma et al. (2023). In this work, we propose a novel perspective for addressing this visual deficiency in vector graphics with an intermediate perception representation.

Image vectorization and program synthesis. Generating vectorized or symbolic representations of visual concepts has been a topic of interest in both the NLP and computer vision communities. Recent work (Vinker et al., 2022; Lee et al., 2023; Ma et al., 2022; Rodriguez et al., 2023; Jain et al., 2023a; Tang et al., 2024; Xing et al., 2024; Hu et al., 2024) has investigated generating vector graphics codes from raster images or text prompts. We focus on the reverse problem of understanding and reasoning about vector graphics as visual inputs. We find that vector graphics reasoning serves as a challenging testbed to evaluate low-level visual reasoning abilities in large multimodal models (LMMs). Although Bubeck et al. (2023); Cai et al. (2023); Zou et al. (2024); Qiu et al. (2024) have demonstrated the potential of text-only large language models (LLMs) in understanding the semantics of vector graphics codes, it remains unclear how to enhance the ability of large *multimodal* models to process *rasterized* vector graphics without access to the underlying code, which is more common in real-world scenarios. Therefore, we propose the intermediate Primal Visual Description representation to further enhance low-level perception and reasoning, without sacrificing the performance of semantic understanding. This work is also heavily inspired by related work in neural-symbolic models (Ritchie et al., 2016; Wu et al., 2017; Yi et al., 2018; Mao et al., 2019; Hsu et al., 2023; Zhang et al., 2023; Trinh et al., 2024). This paradigm aims to de-render visual scenes into structured representations, retrieve programs from the input text, and execute these programs on the image representations. However,

most neural-symbolic work can not be applied to recent LMMs and is limited to specific tasks. We aim to learn a task-agnostic visual description that can be directly reasoned about by off-the-shelf LMMs.

Disentangling perception and reasoning in large multimodal models. Another closely related line of work has investigated disentangling visual perception and reasoning with visual programming (Gupta & Kembhavi, 2023; Surís et al., 2023; Ge et al., 2023; Wu & Xie, 2023) and tool-using (Wu et al., 2023; Liu et al., 2023c; Qiao et al., 2024). These models leverage the code generation capabilities of LLMs to compose and employ a set of vision-language or vision-only models, such as object detection and image caption models, as subroutines for solving visual reasoning tasks. Despite promising performance on natural images, as shown in § 3, we find that these models are still limited by the existing vision-language models’ inability to process low-level primitives effectively.

6 Limitations and Future Work

VDLM represents an initial step to constructing a descriptive intermediate representation for low-level visual reasoning. While our results strongly support this proof of concept, several limitations remain, highlighting directions for future work.

First, due to the scarcity of end-to-end instructional tuning data (discussed in §2.1), the current SVG-to-PVD model is constrained to compositions of simple primitives, for which we can synthesize data without requiring annotation. To enhance PVD’s generalization across more diverse domains, we identify several possible improvements. For instance, incorporating human-created vector graphics with procedural annotations could enrich the ontology and training dataset. Additionally, integrating visual search Wu & Xie (2023) during inference could facilitate the conversion of focused regions or parts into PVD, enabling multi-step perception and reasoning. Second, SVG and PVD are inherently designed for representing 2D vector graphics. Future research is needed to develop a more general intermediate representation that extends beyond 2D vector graphics to encompass 3D structures and natural images.

Acknowledgments

This research is based upon work supported by U.S. DARPA ECOLE Program No. #HR00112390060, AFOSR YIP FA9550-23-1-0127, ONR N00014-23-1-2355, ONR YIP N00014-24-1-2117, ONR MURI N00014-24-1-2748, and the Stanford Institute for Human-Centered AI (HAI). The views and conclusions contained herein are those of the authors and should not be interpreted as necessarily representing the official policies, either expressed or implied, of DARPA, or the U.S. Government. The U.S. Government is authorized to reproduce and distribute reprints for governmental purposes notwithstanding any copyright annotation therein.

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